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RRL Research Task Distribution

Member	Assigned studies	Research focus
Rai	RRL 1 and RRL 4	Job recommendation approaches and experience-based matching
Fiona	RRL 2 and RRL 6	Explainable AI, decision support, and multi-sided fairness
Tine	RRL 3 and RRL 5	Matching algorithms and real-world recruiter evaluation

Required output for each member

Each member should complete **two separate RRL analysis entries** using the same documentation format.

Required column	Information to extract
1. Author(s) and Year	Complete APA 7 reference
2. Research Objective	Problem investigated
3. Algorithm / Model	Methods proposed, implemented, or reviewed
4. Dataset	Source, size, and characteristics
5. Evaluation Metrics	Performance measurements used
6. Key Findings	Actual research results
7. Limitations / Research Gap	Identified weaknesses or limitations
8. Relevance to AYUDA	How the study supports our system
9. Implementation Decision	Adopt, adapt, benchmark, or exclude, with justification

Important: Members should read the actual papers rather than relying solely on abstracts or AI-generated summaries. For systematic reviews, document the methods reviewed rather than claiming the authors implemented all those algorithms.

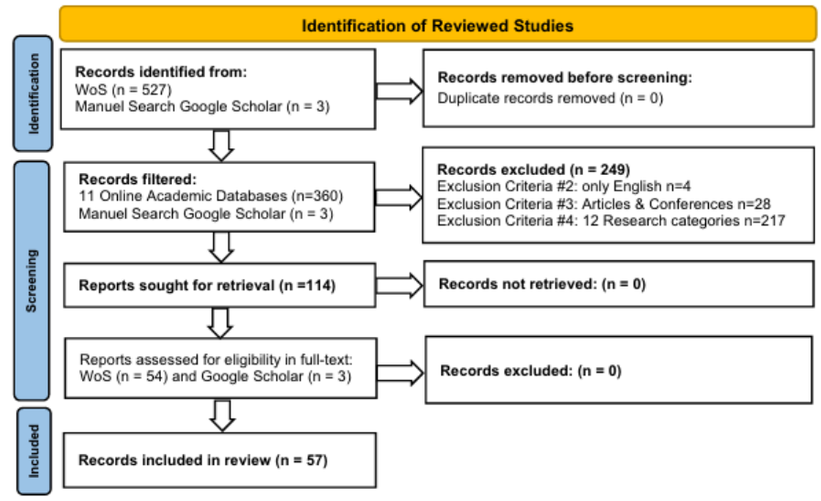
Rai — Recommendation Methods and Career-Based Matching

RRL 1: Job Recommender Systems: A Systematic Literature Review (2025)

Focus on content-based, collaborative, knowledge-based, hybrid, and reciprocal recommendation approaches.

Job recommender systems: a systematic literature review, applications, open issues, and challenges

Author(s) and Year	*Ertuğrul and Bitirim (2025)* 7th Gen APA Citation: Ertuğrul, D. Ç., & Bitirim, S. (2025). Job recommender systems: a systematic literature review, applications, open issues, and challenges. Journal of Big Data, 12(1). https://doi.org/10.1186/s40537-025-01173-y
Research Objective	The study aims to systematically review and critically analyze existing Job Recommender Systems (JRSs), identify commonly used methodologies, classify performance evaluation metrics, examine their applications in Human Resource Management (HRM), and identify research trends, limitations, and gaps.
Algorithm / Model	Collaborative filtering approaches reviewed include item-based recommendation, similarity ranking, fuzzy mathematics, and models incorporating historical interactions and résumé information. Representative studies include Guan et al.'s enterprise–talent recommendation system and Yi

	et al.'s optimized item-based collaborative filtering algorithm.
Dataset	 **Fig. 5** The flow diagram of the article selection process applied in this SLR The study analyzed 57 Job Recommender System research articles and 19 related survey articles published between 2010 and 2023. The reviewed studies utilized different datasets involving candidate profiles, job postings, and historical interactions. As a systematic literature review, the study did not use a single experimental dataset to evaluate all recommendation approaches.
Evaluation Metrics	The study reviews predictive accuracy (MAE, MSE, RMSE), classification-based accuracy (precision, recall, F1-score), ranking-based accuracy (Precision@K, MAP, MRR, NDCG), and non-accuracy metrics (scalability, diversity, coverage).
Key Findings	The study found that collaborative, content-based, hybrid, and knowledge-based filtering are commonly represented in Job Recommender System research, while reciprocal filtering remains comparatively underexplored. Commonly discussed methodologies include similarity-based techniques such as cosine similarity and TOPSIS, machine learning models such as XGBoost and LSTM, and knowledge-based approaches such as ontologies and the Semantic Web. Precision, recall, and F1-score were among the most frequently used evaluation metrics. The reviewed approaches demonstrate different methods of supporting job recommendations, including fuzzy-mathematics-based talent ranking and collaborative filtering that combines historical interactions with résumé information.
Limitations / Research Gap	The review identifies cold-start problems, data sparsity, scalability, interpretability, and data privacy as challenges affecting recommender systems. Collaborative filtering can be particularly constrained by limited historical interaction data, while reciprocal recommendation approaches remain comparatively underexplored in the reviewed literature. *(These are limitations of recommendation approaches and gaps identified across the literature)*
Relevance to AYUDA	The study is relevant to AYUDA's proposed Candidate-to-Job Matching and Explainable DSS because it examines recommendation approaches for matching candidate information with job requirements. Its discussion of recommendation architectures, historical-data limitations, and evaluation methods provides a basis for comparing algorithms according to AYUDA's data availability, matching requirements, and need for understandable recommendations.

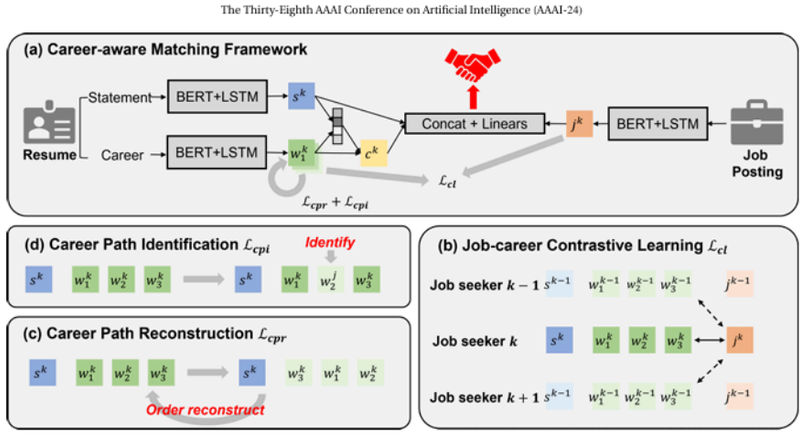
Implement Decision	**Compare / Further Investigate**. The study will serve as a foundation for comparing content-based, collaborative, knowledge-based, hybrid, and reciprocal recommendation approaches. Further investigation will determine which methods are most feasible for AYUDA based on available candidate data, job requirements, explainability, and evaluation criteria.
Synthesis	The reviewed literature establishes different approaches to candidate-job recommendation and identifies the importance of matching accuracy, explainability, and data availability. Further comparison and experimental evidence are required to determine the most appropriate approach for AYUDA.

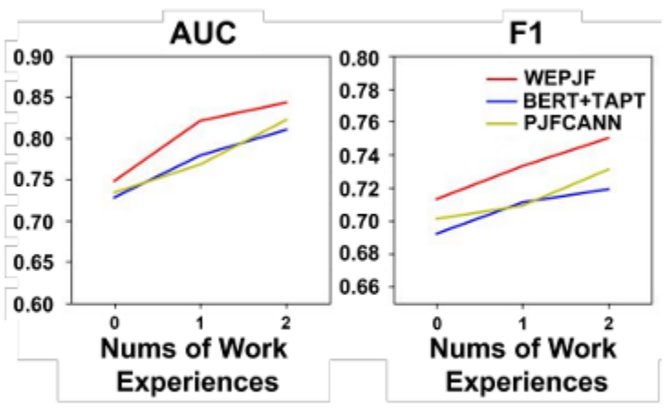
RRL 4: Your Career Path Matters in Person-Job Fit (2024)

Focus on how previous work experience and career progression influence candidate-to-job matching.

Expected contribution: Identify and compare possible matching approaches that AYUDA could use.

Your Career Path Matters in Person-Job Fit

Author(s) and Year	**Gong et al. (2024)** *APA 7th Gen:* **Gong, Z., Song, Y., Zhang, T., Wen, J., Zhao, D., & Yan, R. (2024)**. Your career path matters in Person-Job fit. *Proceedings of the AAAI Conference on Artificial Intelligence*, *38*(8), 8427–8435. https://doi.org/10.1609/aaai.v38i8.28685
Research Objective	The study aims to improve person-job matching by developing a Work Experience Enhanced Person-Job Matching (WEPJM) framework that incorporates candidates' previous work experiences and career-path preferences, specifically consistency, likeness, and continuity, to better assess their suitability for job opportunities.
Algorithm / Model	 Figure 2: Model overview. WEPJM learns representations for input resumes and job postings with hierarchical encoders. The overall objective can be decomposed into the main task of person-job fit and three auxiliary tasks. The auxiliary tasks include (b) *Career Path identification*, (c) *Career Path Reconstruction*, as well as (d) *Job-career Contrastive Learning*. The study proposes the Work Experience Enhanced Person-Job Matching (WEPJM) framework, which uses pre-trained BERT, Bi-LSTM, and an attention mechanism to process résumé statements, chronological work experiences, and job descriptions. It incorporates three auxiliary learning tasks: Career Path Identification (CPI) for consistency, Career Path

	Reconstruction (CPR) for continuity, and Job-Career Contrastive Learning (CL) for likeness. The resulting representations are combined and processed through a Multi-Layer Perceptron (MLP) to predict candidate-job matching scores.
Dataset	The study utilized an anonymized real-world recruitment dataset consisting of 82,362 job postings, 33,285 résumés, and 117,780 work experiences. It included 119,031 positive and 359,721 negative person-job pairs, labeled according to candidate-recruiter interactions and job-viewing behavior on the recruitment platform.
Evaluation Metrics	The study evaluated WEPJM using Accuracy, Precision, Recall, F1-score, and Area Under the Curve (AUC). The proposed model achieved an accuracy of 0.850, precision of 0.844, recall of 0.829, F1-score of 0.837, and AUC of 0.929, outperforming the evaluated baseline models.
Key Findings	 Methods | Acc. | Prec. | Recall | F1 | AUC LR 0.525 0.580 0.523 0.550 0.545 DT 0.624 0.602 0.629 0.615 0.627 NB 0.522 0.472 0.524 0.497 0.530 RF 0.648 0.722 0.629 0.673 0.696 GBDT 0.557 0.612 0.552 0.580 0.594 PJFNN 0.743 0.780 0.693 0.737 0.836 APJFNN 0.760 0.790 0.722 0.750 0.847 JRMPM 0.780 0.799 0.743 0.773 0.870 ResumeGAN 0.774 0.794 0.743 0.767 0.863 PJFCANN 0.813 0.824 0.807 0.816 0.894 BERT 0.768 0.794 0.728 0.760 0.855 BERT+TAPT 0.804 0.819 0.780 0.798 0.887 WEPJM **0.850⋆** **0.844⋆** **0.829⋆** **0.837⋆** **0.929⋆** Table 2: Overall performance of all methods. '⋆' indicates that we accept the improvement hypothesis of our model over the best baseline at a significant test level of 0.01. | Acc. | Prec. | Recall | F1 | AUC All 0.850 0.844 0.829 0.837 0.929 No 0.813 0.818 0.794 0.803 0.889 +CPI 0.827 0.832 0.823 0.828 0.917 +CPR 0.819 0.821 0.815 0.818 0.898 +CL 0.821 0.825 0.821 0.824 0.904 -CPI 0.831 0.833 0.814 0.823 0.913 -CPR 0.842 0.833 0.822 0.827 0.917 -CL 0.834 0.833 0.816 0.825 0.912 Table 3: Ablation studies of auxiliary tasks. We run experiments to test different combinations of the 3 auxiliary tasks, i.e., CPI, CPR, and CL. '+' indicates to use the single auxiliary task only while '-' denotes to exempt the auxiliary task from all three tasks. We enumerate all possible combinations of the auxiliary tasks.  Figure 4: Performance of fresh job seeker (job seekers with less than 3 work experiences). The proposed WEPJM framework outperformed the evaluated baseline models, achieving an accuracy of 0.850, F1-score of 0.837, and AUC of 0.929. Ablation studies demonstrated that incorporating career-path learning tasks improved matching performance, with Career Path Identification (CPI) providing the strongest individual contribution. Case studies further illustrated how considering career progression could produce recommendations more aligned with candidates' professional seniority and development. However, performance declined for candidates with limited work experience.
Limitations / Research Gap	Existing person-job matching approaches have given insufficient attention to candidates' career progression. Although WEPJM addresses this gap, its performance decreases for candidates with limited work experience. Additionally, the study's use of recruitment interactions as positive and negative matching labels may not fully reflect actual candidate qualifications or job suitability.
Relevance to AYUDA	The study is relevant to AYUDA's proposed Candidate-to-Job Matching and Explainable DSS because it demonstrates the importance of considering

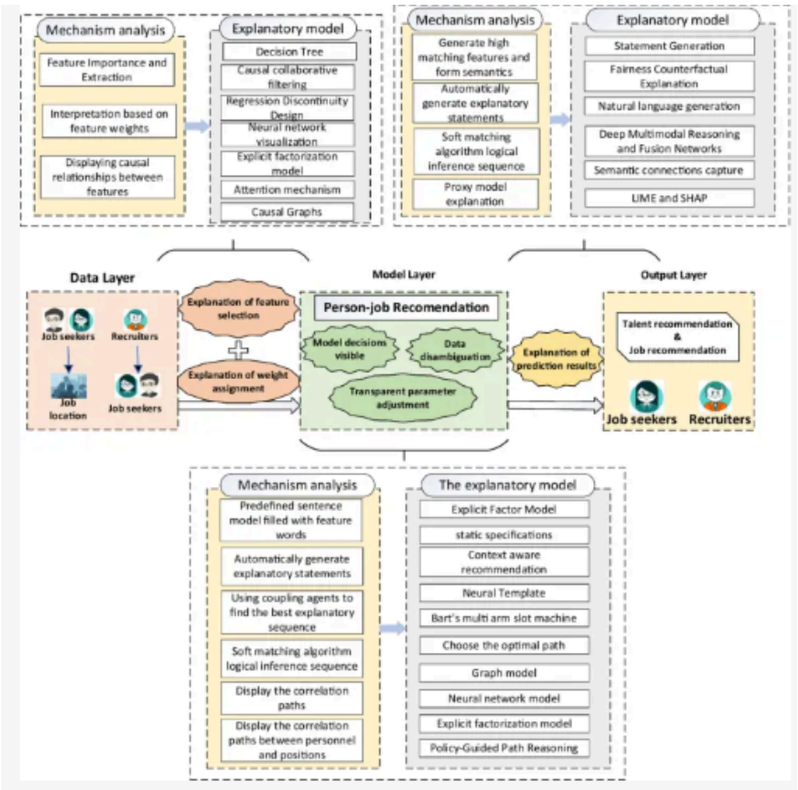
	previous work experiences and career progression when assessing job suitability. Its findings could support incorporating candidates' evolving skills, responsibilities, and professional seniority into the matching process, allowing recruiters to identify opportunities that align more closely with candidates' qualifications and career direction.
Implement Decision	**Adapt**. The study's career-path-aware matching concepts may be incorporated into a simpler, explainable candidate-to-job matching approach rather than implementing the complete WEPJM framework. AYUDA's proposed system could consider verified work experience, career progression, and job requirements as matching criteria while providing understandable explanations for recruiter review. The specific algorithm and criteria will be determined through further research and client validation.
Synthesis	Gong et al. (2024) demonstrate that incorporating career-path information can improve person-job matching performance compared with the evaluated baseline models. Their findings complement earlier research on content-based and explainable recommendations by highlighting the importance of previous work experiences, career progression, and professional seniority. For AYUDA, these concepts may support a simpler, transparent matching approach that considers career history alongside qualifications and job requirements without requiring the complete WEPJM architecture.

Fiona — Explainable DSS and Fairness

RRL 2: Explainable Person–Job Recommendations: Challenges, Approaches, and Comparative Analysis (2025)

Focus on explanation methods, interpretability, and how matching recommendations can be justified.

Explainable person–job recommendations: challenges, approaches, and comparative analysis

Author (s) and Year	Tang, F., Zhu, R., Yao, F., Wang, J., Luo, L., & Li, B. (2025). Explainable person–job recommendations: Challenges, approaches, and comparative analysis. *Frontiers in Artificial Intelligence, 8*, Article 1660548. https://doi.org/10.3389/frai.2025.1660548
Research Objectives	To review explainability methods for person–job recommendation systems and determine how these methods improve transparency, fairness, and user trust compared with black-box approaches. The study specifically examines how recommendations can be understood and justified by job seekers and recruiters.
Algorithm/Model	 The study is a systematic review and does not implement one new algorithm. It reviews existing person–job recommendation models, including content-based, collaborative-filtering, and hybrid approaches. It also organizes explainability methods into three layers: the data layer, which explains feature extraction and weighting; the model layer, which explains how candidate and job information are processed; and the output layer, which explains recommendation results using methods such as SHAP, LIME, natural-language explanations, and counterfactual explanations.
Data Set	The study did not use one original dataset because it was a systematic review. It searched Google Scholar, Web of Science, and CNKI and reviewed English- and Chinese-language studies published from 2019 to August 2025. From 150 screened studies, 85 were included. The reviewed systems commonly used resumes, job descriptions, job requirements, candidate preferences, recruiter preferences, and interaction histories as data sources.
Evaluation Metrics	The paper discusses recommendation measures such as HR@10, Recall@K,

	AUC, and accuracy. In its comparative analysis of six explanation-method families, it used HR@10, fidelity, sparsity, user trust, and the combined Explainability–Performance (E-P) score.
Key Findings	The study found that person–job recommendation systems may become black boxes because users cannot clearly see how data was extracted, how feature weights were assigned, how the model made its decision, or why a recommendation was produced. Explainability can help users understand recommendations, identify possible system problems, support fairness, and improve trust. However, more complex models may provide strong performance while becoming more difficult to interpret.
Limitations / Research Gap	The study was limited to selected databases, studies from 2019–2025, and English or Chinese publications. The researchers also did not pre-register the review. The explanation methods have limitations: some may not fully show the model's reasoning, while others may be difficult to understand, require more data, or need greater computing resources. Many systems also do not adapt to user feedback. Therefore, the study does not identify one explanation method that is best for every recruitment system.
Relevance to AYUDA	The findings support AYUDA's need for transparent candidate-to-job recommendations. AYUDA can show which candidate information was used, which job requirements matched, which requirements were missing, and which information still needs verification. This can help recruiters review and justify recommendations before endorsement.
Implement Decision	**Adapt selected principles rather than implementing every advanced method.** AYUDA may begin with a transparent rule-based or structured explanation layer that displays match-gap breakdowns, visible criteria, source evidence from the candidate profile and job requirements, and recruiter verification. SHAP, LIME, counterfactual explanations, or complex knowledge graphs should only be considered if they are technically feasible and approved.
Synthesis	The study shows that a person–job recommendation system should not only produce a match score. It should also show how the input data was used, how the matching process was performed, and why the recommendation was generated. For AYUDA, this supports an Explainable Decision Support System that gives transparent and reviewable recommendations while keeping the final decision with authorized recruiters and client companies.

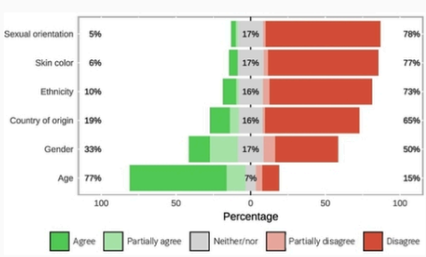
RRL 6: Mapping Stakeholder Needs to Multi-Sided Fairness in Candidate Recommendation for Algorithmic Hiring (2025)

Focus on fairness and the different needs of applicants, recruiters, and client companies.

Expected contribution: Determine how AYUDA's proposed DSS could provide transparent and responsible recommendations.

Mapping Stakeholder Needs to Multi-Sided Fairness in Candidate Recommendation for Algorithmic Hiring

Author(s) and Year	Kaya, M., & Bogers, T. (2025). Mapping stakeholder needs to multi-sided fairness in candidate recommendation for algorithmic hiring. n *RecSys '25: Proceedings of the Nineteenth ACM Conference on Recommender Systems* (pp. 257–267). Association for Computing Machinery. https://doi.org/10.1145/3705328.3748079

Research Objectives	To identify and map fairness metrics suitable for a multi-stakeholder candidate recommendation scenario and determine how these metrics can support the development of fairness-aware machine-learning systems. The study also aims to map and incorporate understanding and the lived experience of stakeholders into fairness metric design, although we consider multiple different stakeholders.
Algorithm/Model	The study used semi-structured interviews with 40 stakeholders and thematic analysis to identify fairness concerns in algorithmic hiring. It describes an existing content-based candidate recommendation system using a cross-encoder model to match CVs with job postings and rank candidates. Rather than implementing a new fairness-aware algorithm, the study mapped stakeholder needs to existing fairness metrics and proposed their use in future auditing and fairness-aware system development.
Data Set	The study's algorithmic hiring scenario used Jobindex's CV database, which contained more than 160,000 candidate CVs. The system also used job-posting text to calculate candidate–job matching scores. For the qualitative study, the researchers collected data from 40 stakeholder interviews and a screening survey with 246 job seekers. However, they did not analyze an actual recommendation-outcome dataset or publicly release a machine-learning dataset.
Evaluation Metrics	The study mapped stakeholder concerns to possible fairness metrics, including Conditional Demographic Parity, rank-aware fairness, diversity, coverage, the Gini index, and company parity. These metrics may be used to examine whether qualified candidates and companies receive fair opportunities. However, the researchers did not calculate these metrics using actual recommendation outcomes. They proposed them for future algorithmic auditing and fairness-aware system development.
Key Findings	Figure 1:  Experiences with unfairness of the participants in the screener survey used to recruit job seekers ($N = 246$). All job seekers that had experienced discrimination or unfair treatment while job seeking ($n = 74$) were asked for the suspected cause: age, gender, country of origin, ethnicity, skin color, and/or sexual orientation. The screening survey showed that 74 out of 246 job seekers, or 30.1%, reported experiencing discrimination or unfair treatment while seeking employment. Among these respondents, age was the most commonly suspected factor, followed by gender. This finding highlights why fairness should be monitored in algorithmic hiring systems. However, the survey did not prove that the unfair treatment was caused by the recommendation algorithm. During the interviews, many participants emphasized that candidates should first meet the qualifications stated in the job advertisement. After considering qualifications, many participants believed that candidates should be treated equally regardless of other personal characteristics. The study also found that fairness should consider multiple stakeholders, including job seekers, recruiters, companies,

	and recruitment agencies.
Limitations / Research Gap	The study identified that fairness is difficult to quantify because there is no single definition or universally correct fairness metric. Although the researchers mapped stakeholder perspectives to possible fairness metrics, they did not empirically test these metrics using actual candidate recommendation outcomes. The study was also limited by the lack of diversity among the participating job seekers and companies. In addition, most participants came from Denmark and were connected to one recruitment agency, so the findings may not fully apply to other countries, cultures, or recruitment systems. These limitations show the need for future research using real recruitment data and more diverse stakeholders.
Relevance to AYUDA	This study is relevant to AYUDA because it emphasizes the need for fair and transparent candidate recommendation systems. Its focus on qualifications, fairness metrics, stakeholder needs, and algorithmic auditing can guide AYUDA in developing a DSS that recommends suitable candidates while reducing bias. AYUDA can apply a qualification-first approach, provide explanations for recommendations, monitor whether qualified candidates are fairly considered, and maintain human recruiter review before final decisions are made. This supports a more responsible and multi-stakeholder hiring process for applicants, recruiters, and client companies.
Implement Decision	**Adopt and adapt.** AYUDA can adopt the study's multi-stakeholder fairness approach and adapt its proposed metrics for system auditing. However, the metrics should be tested using AYUDA's own candidate and recruitment data before full implementation.
Synthesis	Kaya and Bogers (2025) emphasized that algorithmic hiring systems should consider fairness for multiple stakeholders, including job seekers, recruiters, companies, and recruitment agencies. Their study identified fairness metrics for evaluating qualified candidates and company access, but it did not implement or test a fairness-aware recommendation algorithm using actual hiring outcomes. In comparison, AYUDA can apply these insights by developing a qualification-based and transparent candidate recommendation DSS with fairness monitoring, explanations, and human recruiter review. Thus, the study provides a foundation for making AYUDA's recommendations more responsible and fair.

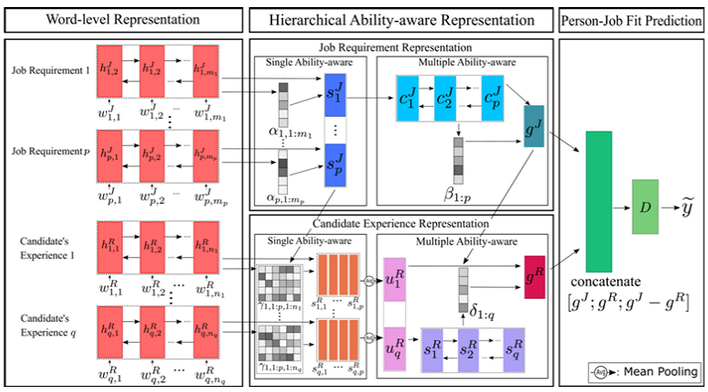
Tine — Matching Algorithms and Practical Evaluation

RRL 3: Enhancing Person-Job Fit for Talent Recruitment (2018)

Focus on the Ability-aware Person-Job Fit Neural Network (APJFNN), its architecture, algorithms, and evaluation results.

Enhancing Person-Job Fit for Talent Recruitment:An Ability-aware Neural Network Approach

Author(s) and Year	*Qin, C., Zhu, H., Xu, T., Zhu, C., Jiang, L., Chen, E., & Xiong, H. (2018)*. Enhancing person-job fit for talent recruitment: An ability-aware neural network approach. In Proceedings of the 41st International ACM SIGIR Conference on Research & Development in Information Retrieval (pp. 25–34). Association for Computing Machinery

Research Objective	To improve Person-Job Fit prediction by automatically analyzing job requirements and candidate experiences, reducing dependence on manual feature evaluation, and providing more interpretable recruitment matching results.
Algorithm / Model	 **Figure 2: An illustration of the proposed Ability-aware Person-Job Fit Neural Network (APJFNN), which can be separated into three components, namely Word-level Representation, Hierarchical Ability-aware Representation and Person-Job Fit Prediction. Meanwhile, two different hierarchical structures are used to learn the ability-aware representation of job requirement and candidate experience respectively.** The study proposed the **Ability-aware Person-Job Fit Neural Network (APJFNN)**, which uses **Bidirectional Long Short-Term Memory (BiLSTM)** networks for semantic text representation and four hierarchical ability-aware attention mechanisms to identify important job requirements and relevant candidate experiences. A fully connected prediction network estimates Person-Job Fit.
Dataset	The researchers used anonymized historical recruitment records from a high-technology company in China. The original dataset contained 17,766 job postings and 898,914 resumes. After preprocessing, the retained dataset included 3,652 job postings, 533,069 resumes, 12,796 successful applications, and 1,058,547 failed applications. The data covered technology, product, user interface, and other job categories.
Evaluation Metrics	Accuracy, Precision, Recall, F1-score, and Area Under the ROC Curve (AUC). The researchers also evaluated performance across different training-data proportions and examined computational efficiency.
Key Findings	APJFNN outperformed the evaluated baseline models, achieving 75.59% accuracy, 75.45% precision, 76.03% recall, 75.74% F1-score, and 0.8316 AUC on the balanced evaluation dataset. The hierarchical attention mechanisms improved predictive performance compared with the basic RNN-based model (BPJFNN) and helped highlight important requirements and relevant candidate experiences.
Limitations / Research Gap	The study relied on historical recruitment outcomes from a single company, which may limit generalizability. Its original dataset was highly imbalanced, requiring undersampling for evaluation. The model also requires substantial labeled recruitment data and a relatively complex neural architecture. Additionally, the study did not establish whether its attention-based explanations faithfully represent all factors influencing predictions or whether the model performs fairly across different applicant groups.
Relevance to AYUDA	The study supports AYUDA's proposed candidate-to-job matching and candidate rediscovery features by demonstrating how semantic analysis can identify relationships between job requirements and candidate qualifications beyond exact keyword matching. Its attention-based approach also provides a conceptual basis for highlighting relevant skills and experiences, helping recruiters understand candidate recommendations.
Implement Decision	Adapt and Benchmark. AYUDA can adapt the study's concepts of semantic matching, requirement weighting, and explainable candidate recommendations without necessarily implementing the complete APJFNN architecture. The proposed matching approach should be evaluated against simpler baseline methods using AYUDA's available recruitment data before selecting a final model.

Synthesis	Qin et al. (2018) demonstrated the potential of deep learning in improving Person-Job Fit prediction through their proposed Ability-aware Person-Job Fit Neural Network (APJFNN). By integrating Bidirectional Long Short-Term Memory (BiLSTM) networks with four hierarchical attention mechanisms, the model analyzes the semantic relationships between job requirements and candidate experiences while identifying the qualifications most relevant to a position. The study achieved an accuracy of 75.59%, an F1-score of 75.74%, and an AUC of 0.8316, outperforming the evaluated baseline models. These findings suggest that ability-aware semantic matching can improve recruitment predictions while providing greater interpretability through the identification of important requirements and relevant experiences. The study provides a valuable foundation for AYUDA's proposed candidate-to-job matching and candidate rediscovery features. Its approach demonstrates how recruitment systems can move beyond traditional keyword matching by considering the contextual relevance of candidate qualifications. However, the model's dependence on extensive historical recruitment data, its computational complexity, and its limited evaluation within a single organization highlight the need for adaptation. Therefore, AYUDA can adopt the principles of semantic matching and explainable recommendations while evaluating simpler and more practical approaches suited to its available data and operational requirements.

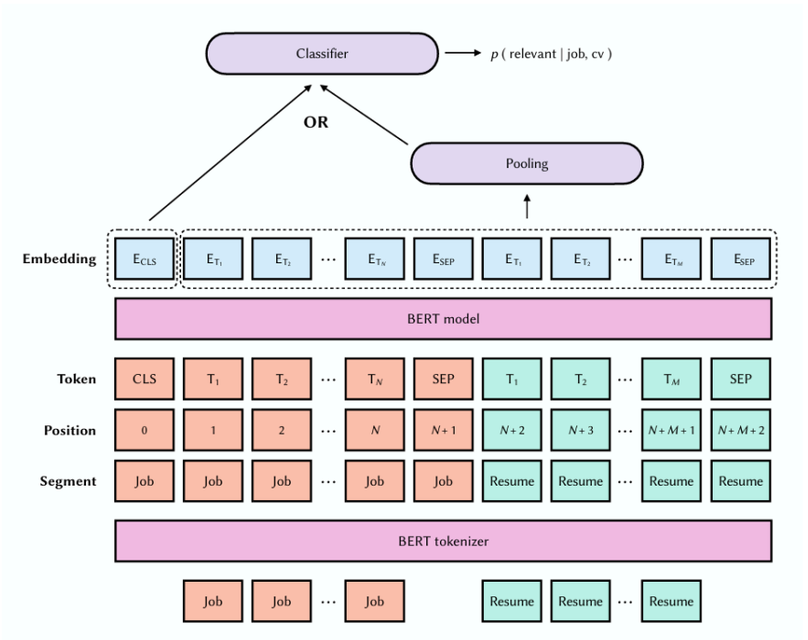
RRL 5: Analyzing the Effects of a Human-in-the-Loop Candidate Recommendation Algorithm at Jobindex (2026)

Focus on SPBERT, human-in-the-loop recommendations, recruiter adoption, and real-world evaluation.

Expected contribution: Assess the technical feasibility of matching models and how recommendation performance should be measured.

Analyzing the Effects of a Human-in-the-Loop Candidate Recommendation Algorithm at Jobindex

Author(s) and Year	Kaya, M., & Bogers, T. (2027). Analyzing the effects of a human-in-the-loop candidate recommendation algorithm at Jobindex. ACM Transactions on Recommender Systems, 5(2), Article 26. https://doi.org/10.1145/3801092
Research Objective	To develop, deploy, and evaluate an AI-based candidate recommendation system that supports recruiters in identifying suitable candidates while maintaining human oversight. The study also investigated how recruiter engagement with AI-generated recommendations affects recruitment productivity, candidate response rates, and search effort in real-world operations.

Algorithm / Model	 Fig. 5. Overview of the SPBERT architecture. The researchers implemented **Sentence-pair BERT (SPBERT)**, a Transformer-based model that jointly processes job postings and resumes to predict candidate relevance. The model uses a pre-trained BERT encoder, a multilayer perceptron (MLP) classification layer, and binary cross-entropy loss for training. It was compared with Job Title-based Recommendation with Item History (JTIH), Person-Job Fit Neural Network (PJFNN), Basic Person-Job Fit Neural Network (BPJFNN), and Sentence-BERT (SBERT). The deployed system followed a **Human-in-the-Loop (HITL) approach**, allowing recruiters to review recommendations and make final shortlisting decisions.
Dataset	The offline experiments used historical recruitment data from Jobindex, Denmark, covering April 2018 to May 2022. The dataset contained approximately 138,000 job postings, 120,941 CVs, and 2,427,288 job–resume pairs. For the online A/B test, 38 recruiters processed 1,334 job postings over 25 days in March 2023. The subsequent deployment analysis covered 41,390 recruitment tasks over 17 months (April 2023–September 2024), examining recruiters' interactions with recommended candidates and their responses.
Evaluation Metrics	Offline evaluation used Area Under the ROC Curve (AUC) for classification and Normalized Discounted Cumulative Gain at 20 (NDCG@20) for candidate ranking. Online evaluation used Positive Response Rate (PRR) and Contacted From Recommendations (CFR). The longitudinal analysis additionally measured Negative Response Rate (NRR), number of contacted candidates, positive and negative responses, number of search queries, and effort required to identify relevant candidates.
Key Findings	SPBERT outperformed the evaluated baseline models in offline experiments, achieving 0.946 AUC and 0.186 NDCG@20. In the online A/B test, SPBERT achieved a PRR of 13.5% and CFR of 41.1%, compared with JTIH's 12.7% PRR and 24.75% CFR. The CFR improvement was statistically significant, while the PRR difference was not. During the 17-month deployment, recruiters who engaged with recommendations achieved a 12.72% higher PRR, contacted 15.92% more candidates, and submitted 22.73% fewer search queries than those who skipped recommendations. These findings indicate that AI-assisted recommendations can support recruitment efficiency when integrated into existing workflows.
Limitations / Research Gap	The study identified challenges involving recruiter distrust of AI, inconsistent adoption, and insufficient explanations for recommended candidates. Its skill and occupation highlighting mechanism did not always provide satisfactory evidence of candidate relevance. The study also acknowledged possible self-selection bias in recruiter feedback and limitations in categorizing engagement into only two groups: Check and Skip. Furthermore, the system

	was evaluated within Jobindex's recruitment environment, which may limit generalizability to other organizations. The longitudinal comparison was observational, so differences between Check and Skip groups should not automatically be interpreted as causal effects.
Relevance to AYUDA	The study supports AYUDA's proposed AI-assisted candidate recommendation, semantic job matching, and candidate rediscovery features. SPBERT demonstrates how job postings and resumes can be jointly analyzed to identify relevant candidates despite differences in terminology and document structure. Its HITL approach also supports AYUDA's objective of assisting recruiters without replacing their professional judgment. Additionally, the findings on skill highlighting, recruiter trust, and workflow integration provide a basis for designing explainable recommendations that recruiters can review before shortlisting candidates.
Implement Decision	Adapt and Benchmark. AYUDA can adapt the study's semantic matching and Human-in-the-Loop principles by incorporating AI-generated candidate recommendations into the existing recruitment workflow while retaining recruiter control over final decisions. SPBERT should be considered a candidate model for benchmarking against simpler semantic matching approaches using AYUDA's available recruitment data. The system should also prioritize relevant skill highlighting, recommendation explanations, and recruiter feedback to improve usability and trust.
Synthesis	Kaya and Bogers (2027) investigated the development and deployment of a Human-in-the-Loop candidate recommendation system at Jobindex using Sentence-pair BERT (SPBERT). The model jointly analyzes job postings and resumes to predict candidate relevance, addressing limitations of traditional keyword matching and structured-input neural models. SPBERT achieved an AUC of 0.946 and NDCG@20 of 0.186, outperforming the evaluated baseline algorithms. The 17-month deployment analysis further showed that recruiters who engaged with AI-generated recommendations achieved higher positive response rates, contacted more candidates, and performed fewer search queries. However, the study identified challenges involving recruiter trust, recommendation explainability, and inconsistent system adoption. The findings provide a methodological foundation for AYUDA's proposed candidate-to-job matching and candidate rediscovery features. In particular, the study demonstrates the potential of Transformer-based semantic matching to process recruitment information while emphasizing the importance of maintaining human oversight. For AYUDA, the proposed approach can be adapted by integrating explainable candidate recommendations into recruiters' existing workflows and benchmarking semantic matching models against available recruitment data. This supports the development of an AI-assisted recruitment system that improves candidate identification without removing recruiters' responsibility for final decisions.